Parcial 1

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Problema 2

$$\mathbf{E}(\mathbf{A}|x_Q) = \mathbf{E}_H(\mathbf{A}) + \mathbf{E}_Q(\mathbf{A}|x_Q)$$

donde $\mathbf{E}_H$ y $\mathbf{E}_Q$ son los campos eléctricos generados por el hilo y la carga puntual, respectivamente, y $\mathbf{A} = (0.2, 0, 0)$.

$$\begin{aligned}
\mathbf{E}_H(\mathbf{A}) &= \frac{1}{4\pi\epsilon_0}\int_H \frac{\lambda}{\|\mathbf{A}-\mathbf{r}_H\|^3}(\mathbf{A}-\mathbf{r}_H).ds && (1)\\
&= \frac{1}{4\pi\epsilon_0}\int_{-0.1}^{0.1} \frac{\lambda}{\|\mathbf{A}-\phi(t)\|^3}(\mathbf{A}-\phi(t))\,\|\phi'(t)\|\,dt && (2)\\
&= \frac{1}{4\pi\epsilon_0}\int_{-0.1}^{0.1} \frac{\lambda}{(0.2^2+t^2)^{3/2}}(0.2,-t,0)\,dt && (3)\\
&= \frac{\lambda}{4\pi\epsilon_0}\left(\frac{t}{0.2\sqrt{0.2^2+t^2}}\Big|_{-0.1}^{0.1}, \frac{1}{\sqrt{0.2^2+t^2}}\Big|_{-0.1}^{0.1}, 0\right) && (4)\\
&= \frac{\lambda}{4\pi\epsilon_0}\left(\frac{0.2}{0.2\sqrt{0.2^2+0.1^2}}, 0, 0\right)\\
&= \frac{\lambda}{4\pi\epsilon_0}\frac{10}{\sqrt{5}}\hat{\mathbf{x}} = 0.8042\,\hat{\mathbf{x}}\,N/\mu C\\
\mathbf{E}_Q(\mathbf{A}|x_Q) &= \frac{1}{4\pi\epsilon_0}\frac{Q}{(0.2-x_Q)^2}\,\text{sign}(0.2-x_Q)\hat{\mathbf{x}} && (5)
\end{aligned}$$

Notas:

1. Eq. 1 es la definición de la ley de Coulomb para una distribución de carga lineal.

2. Eq. 2 sigue de la definción de la integral de linea para una parametrización ϕ del hilo H:

$$\begin{aligned} \phi : [-0.1, 0.1] &\to H \\ \phi(t) &= (0, t, 0) \\ \phi'(t) &= (0, 1, 0) \end{aligned}$$

3. En la Eq. 3 usamos:

$$\begin{aligned} \mathbf{A} - \phi(t) &= (0.2, 0, 0) - (0, t, 0) = (0.2, -t, 0) \\ \|\mathbf{A} - \phi(t)\|^3 &= (0.2^2 + t^2)^{3/2} \\ \|\phi'(t)\| &= 1 \end{aligned}$$

   entonces:

$$\frac{1}{4\pi\epsilon_0} \int_{-0.1}^{0.1} \frac{\lambda}{\|\mathbf{A} - \phi(t)\|^3} (\mathbf{A} - \phi(t)) \, \|\phi'(t)\| \, dt = \frac{1}{4\pi\epsilon_0} \int_{-0.1}^{0.1} \frac{\lambda}{(0.2^2 + t^2)^{3/2}} (0.2, -t, 0) \, dt$$

4. En la Eq. 4 usamos las siguientes primitivas de una tabla de integrales:

$$\begin{aligned} \int \frac{x}{(x^2 + t^2)^{3/2}} dt &= x \frac{1}{x^2} \frac{t}{\sqrt{x^2 + t^2}} = \frac{1}{x} \frac{t}{\sqrt{x^2 + t^2}} \\ \int \frac{t}{(x^2 + t^2)^{3/2}} dt &= -\frac{t}{\sqrt{x^2 + t^2}} \end{aligned}$$

Parte a

$$\begin{aligned} \mathbf{E}(\mathbf{A}|x_Q = 0.4) &= \mathbf{E}_H(\mathbf{A}) + \mathbf{E}_Q(\mathbf{A}|x_Q = 0.4) \\ &= 0.8042 \, \hat{\mathbf{x}} - 1.7984 \, \hat{\mathbf{x}} = -0.9941 \, \hat{\mathbf{x}} \, \mathrm{N}/\mu C \end{aligned}$$

Parte b

Tengo que encontrar un x_Q tal que $\mathbf{E}(\mathbf{A}|x_Q) = 0$ o $\mathbf{E}_Q(\mathbf{A}|x_Q) = -\mathbf{E}_H(\mathbf{A}) = -0.8042\ \hat{\mathbf{x}}\ \mathrm{N}/\mu C$. Usando la Eq. 5 esto es

$$-\frac{1}{4\pi\epsilon_0}\frac{Q}{(0.2 - x_Q)^2} = -0.8042 \tag{6}$$

Resolviendo la Eq. 6 obtengo

$$x_Q = 0.2 + \sqrt{\frac{1}{4\pi\epsilon_0}\frac{Q}{0.8042}} = 0.5\ \mathrm{m}$$

Parte c

Llamemos $\mathbf{D} = (d, 0, 0)$ al punto donde se encuentra la carga Q. Sabemos que $W = q_2(V(\mathbf{C}) - V(\mathbf{B}))$. Por la definición de potencial:

$$\begin{aligned} V(\mathbf{C}) - V(\mathbf{B}) &= -\int_{\mathbf{B}}^{\mathbf{C}} \mathbf{E}.ds = -\int_{\mathbf{B}}^{\mathbf{C}} (\mathbf{E}_H + \mathbf{E}_Q).ds = -\left(\int_{\mathbf{B}}^{\mathbf{C}} \mathbf{E}_H.ds + \int_{\mathbf{B}}^{\mathbf{C}} \mathbf{E}_Q.ds\right) \\ &= (V_H(\mathbf{C}) - V_H(\mathbf{B})) + (V_Q(\mathbf{C}) - V_Q(\mathbf{B})) \end{aligned}$$

Por simetría, $V_H(\mathbf{C}) = V_H(\mathbf{B})$. Por otro lado, sabemos que para una carga aislada se cumple que:

$$V_Q(\mathbf{C}) - V_Q(\mathbf{B}) = \frac{1}{4\pi\epsilon_0} Q\left(\frac{1}{\|\mathbf{D} - \mathbf{C}\|} - \frac{1}{\|\mathbf{D} - \mathbf{B}\|}\right)$$

entonces

$$\begin{aligned} W = q_2\left(V(\mathbf{C}) - V(\mathbf{B})\right) \ &= q_2\left(V_Q(\mathbf{C}) - V_Q(\mathbf{B})\right) = \frac{1}{4\pi\epsilon_0} q_2 Q\left(\frac{1}{\|\mathbf{D} - \mathbf{C}\|} - \frac{1}{\|\mathbf{D} - \mathbf{B}\|}\right) \\ &= 0.5755\ \mathrm{N\,m} \end{aligned}$$